

🧩 COMPLETE DBMS PROJECT BUILD-UP (Bottom-Up)

🟢 PHASE 1 — Schema Creation (Foundation)



🟡 PHASE 2 — Core Transactional Aggregations (Required Before Analytics)

Dashboard Implementations

1. Daily Sales Dashboard

File: `DailySalesDashboard.jsx`

Features:

- Year-based filtering (All Years or specific year)
- Summary statistics cards showing:
 - Total Sales
 - Total Orders
 - Items Sold
 - Average Daily Sales
- Interactive visualizations
- Detailed data table with average order value calculation

Charts Used:

1. **Line Chart** - Daily Sales Trend
 - Type: Line chart with area gradient
 - Shows sales amount over time
 - Custom tooltip with formatted currency
 - Formatted Y-axis with K/M notation
2. **Line Charts (Side-by-Side)** - Daily Orders & Items Sold
 - Two separate line charts in grid layout
 - Left: Daily Orders (green gradient)
 - Right: Items Sold (yellow gradient)
 - Gradient fills under lines for visual appeal
 - Interactive tooltips

Data Table Columns:

- Date
 - Sales Amount
 - Total Orders
 - Items Sold
 - Average Order Value (calculated)
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2. Quantity Sold Dashboard

File: `QuantitySoldDashboard.jsx`

Features:

- Multiple view types with tab navigation
- Dynamic summary cards that change based on view
- Year filtering for specific views
- Modern gradient-styled buttons

Charts Used:

1. **Bar Chart (Vertical)** - Per Product View
 - Top 20 products by quantity sold
 - Gradient fill (purple to violet)
 - Rounded bar corners
 - Rotated X-axis labels
2. **Bar Chart (Vertical)** - Per Year View
 - Top 10 products for selected year
 - Year dropdown selector
 - Green gradient fill
 - Custom tooltips
3. **Pie Chart** - Per Category View
 - Distribution of quantity sold by category
 - 7-color palette
 - Percentage labels on slices
 - Custom legend with hover effects
 - Interactive color-coded category list
4. **Bar Chart (Vertical)** - Per Category & Year View
 - Top 10 categories for selected year
 - Year dropdown selector
 - Orange/yellow gradient fill

View Types:

- Per Product (all-time top performers)
- Per Year (year-specific product analysis)
- Per Category (category distribution)
- Per Category & Year (category performance by year)

Summary Cards:

- Total Quantity Sold
 - Total Products/Categories (dynamic label)
 - Average Quantity
 - Top Performer
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3. Revenue Per Product Dashboard

File: `RevenuePerProductDashboard.jsx`

Features:

- Year filtering (All Years or specific year)
- Revenue-focused analytics

- Comprehensive data breakdown

Charts Used:

1. **Horizontal Bar Chart** - Revenue by Product
 - All products ranked by revenue
 - Color: Blue gradient
 - Dynamic height based on data count
 - Formatted currency on X-axis (K/M notation)
 - Custom tooltips with product details

Summary Cards:

- Total Revenue
- Total Products
- Average Revenue per Product
- Top Product

Data Table Columns:

- Rank
 - Product Name
 - Year (conditional - only shown when year selected)
 - Total Revenue
 - Revenue Share (percentage)
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4. Revenue Per Seller Dashboard

File: `RevenuePerSellerDashboard.jsx`

Features:

- Seller performance analysis
- Year-based filtering
- Multiple visualization perspectives

Charts Used:

1. **Horizontal Bar Chart** - Revenue by Seller
 - All sellers ranked by revenue
 - Color: Blue
 - Dynamic height adjustment
 - Currency formatting
2. **Horizontal Bar Chart** - Products Sold by Seller

- Quantity metrics per seller
- Color: Green
- Number formatting with commas

Summary Cards:

- Total Sellers
- Total Revenue
- Total Products Sold

Data Table Columns:

- Rank
 - Seller Name
 - Revenue
 - Products Sold
 - Average Revenue per Product (calculated)
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5. Revenue Per Category Dashboard

File: `RevenuePerCategoryDashboard.jsx`

Features:

- Category-level revenue analysis
- Multiple chart types for comprehensive view
- Year filtering capabilities

Charts Used:

1. **Pie Chart** - Revenue Distribution
 - Percentage-based revenue allocation
 - 8-color palette
 - Percentage labels on slices
 - Interactive legend
2. **Pie Chart** - Products Sold Distribution
 - Quantity distribution by category
 - Matches revenue chart styling
 - Dual-chart layout for comparison
3. **Horizontal Bar Chart** - Revenue by Category
 - Linear representation of category revenue
 - Blue gradient fill
 - Formatted currency axis
4. **Horizontal Bar Chart** - Products Sold by Category

- Quantity metrics per category
- Green gradient fill
- Number formatting

Summary Cards:

- Total Categories
- Total Revenue
- Total Products Sold

Data Table Columns:

- Rank
 - Category Name
 - Revenue
 - Products Sold
 - Average Revenue per Product (calculated)
 - Revenue Share (percentage)
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PHASE 3 — Time-Based & Customer-Based Metrics

- 16. **Monthly Revenue per Year**
 - 17. **Monthly Order Count (Has to be sorted using all years, also per year)**
 - 18. **Monthly Sales Trend (Using 16, 17)**
 - 19. **Average Order Value (AOV)**
 - 20. **Customer Lifetime Value (CLTV)**
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PHASE 4 — Advanced Ranking & Segmentation

- 21. **Inactive Sellers Detection**
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PHASE 5 — Returns, Loss & Risk Analysis

- 26. **Most Returned Products (All time, per year, per month(another tab))**
- 27. **Return Rate per Product (All time, per year, per month(another tab))**
- 28. **Revenue Loss Due to Returns (All time, per year, per month(another tab))**

● PHASE 6 — Functions (Reusable Analytics Layer)

29. Monthly Sales Summary Function

- Function (RETURNS TABLE)

30. Customer CLTV Function

- Function

31. Product Profit Margin Function

- Function

32. Seller Performance Function

- Function

33. Revenue Decrease Ratio (YoY) Function

- Function

● PHASE 7 — Procedures (Controlled Data Mutation)

34. Create Order Procedure

- Procedure (transactional)

35. Add Order Items Procedure

- Procedure

36. Update Inventory Procedure

- Procedure

37. Process Return Procedure

- Procedure

38. Mark Order as Shipped Procedure

- Procedure
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PHASE 8 — Triggers (Automation & Integrity)

39. Auto Deduct Inventory on Sale

- Trigger (**AFTER INSERT** on order_items)

40. Prevent Order if Stock Insufficient

- Trigger (**BEFORE INSERT**)

41. Auto Update Order Status on Payment Success

- Trigger

42. Auto Restore Inventory on Return

- Trigger

43. Auto Mark Order Completed

- Trigger (payment + shipping)
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PHASE 9 — Optimization & Intelligence (Exceptional)

44. Low Stock Alert View

- View / Materialized View

45. Fraud Pattern Detection Query

- Query

46. Revenue Drop Detection Query

- Query

47. Daily Inventory Snapshot Job

- Scheduled Procedure